TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	57.3 ± 5.24
Correlation Stability	78.5 ± 2.20
Distribution Consistency	62.4 ± 7.14
Entropy Preservation	48.3 ± 4.08
Mean Z Score Drift	0.6 ± 0.07
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	100.0 ± 0.00
Feature Accuracy	61.6 ± 5.62
Novelty Score	0.9 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: hepatitis.csv

1. Dataset Portrait

Attribute	Value
Rows	155
Columns	20
Numeric features	20
Categorical features	0
Missing cells (%)	5.4%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	41.200	44.218	+3.018	7.3%	OK
Sex	1.103	1.000	-0.103	9.4%	OK
Steroid	1.506	1.548	+0.042	2.8%	OK
Antivirals	1.845	2.000	+0.155	8.4%	OK
Fatigue	1.351	1.161	-0.189	14.0%	Warn
Malaise	1.604	1.516	-0.088	5.5%	OK
Anorexia	1.792	1.782	-0.010	0.6%	OK
Liver Big	1.828	1.855	+0.027	1.5%	OK
Liver Firm	1.583	1.726	+0.142	9.0%	OK
Spleen Palpable	1.800	2.000	+0.200	11.1%	Warn
Spiders	1.660	1.823	+0.163	9.8%	OK
Ascites	1.867	2.000	+0.133	7.1%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	57.3 ± 5.24
Correlation Stability (*2)	78.5 ± 2.20
Distribution Consistency (*3)	62.4 ± 7.14
Entropy Preservation (*4)	48.3 ± 4.08
Mean Z Score Drift (*5)	0.6 ± 0.07
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	100.0 ± 0.00
Feature Accuracy (*10)	61.6 ± 5.62
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7903225806451613	0.7129032258064516	0.8129032258064516	0.022580645161290314	
F1	0.38948306595365423	0.31387566137566136	0.4108683473389356	0.02138528138528138	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7903225806451613	0.7967741935483871	0.832258064516129	0.04193548387096774	
F1	0.561449871078354	0.5231529804904418	0.6207002801120448	0.05925040903369076	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.864516129032258	0.7387096774193549	0.8483870967741934	-0.016129032258064512	
F1	0.5893407761828813	0.4488484848484883	0.5385658785658786	-0.05077489761700288	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.867741935483871	0.8	0.8516129032258064	-0.016129032258064496	
F1	0.6104539320328793	0.5052014652014651	0.5679642579642579	-0.04248967406862143	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7806451612903226	0.6516129032258065	0.8096774193548386	0.029032258064516113	
F1	N/A	0.3087651727357609 3	0.02	0.02	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.8225806451612903	0.8419354838709676	0.003225806451612915	
F1	0.5718392134181608	0.5892344497607656	0.518092463092463	-0.05374675032569769	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7903225806451613	0.7774193548387097	0.8225806451612903	0.03225806451612904	
F1	0.46331890331890335	0.4865656565656553	0.5476217573276397	0.08430285400873636	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8419354838709676	0.8	0.8419354838709676	-2.2204460492503132e-17	
F1	0.517094017094017	0.34716374269005845	0.5461904761904761	0.02909645909645909	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8064516129032258	0.6677419354838710	0.8064516129032258	0	
F1	N/A	0.12375	N/A	0	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8548387096774194	0.7612903225806452	0.832258064516129	-0.022580645161290325	
F1	0.5980719280719281	0.469075924075924	0.5191863691863692	-0.07888555888555888	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
2	50.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0	2.0
3	78.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0
5	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
8	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
9	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
10	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
11	39.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0
12	32.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
13	41.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
14	30.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
15	47.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
16	38.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
17	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
18	40.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
19	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
20	38.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	72.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
2	72.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
3	7.0	1.0	1.0	2.0	2.0	1.0	2.0	2.0	1.0	2.0
4	72.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
5	7.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	72.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
7	61.0	1.0	1.0	2.0	1.0	1.0	2.0	nan	nan	2.0
8	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
9	30.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
10	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	78.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	78.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0	2.0
3	7.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
4	78.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
5	7.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
6	78.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
7	60.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	1.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
9	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	36.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	78.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	7.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	7.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	7.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
5	7.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0
6	7.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
7	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	42.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	78.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
2	20.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
3	78.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	20.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
5	20.0	1.0	2.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0
6	20.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
9	25.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
10	38.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	78.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	7.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
3	78.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
4	7.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
5	7.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	78.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
7	30.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	45.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	36.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	78.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
2	7.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0
3	7.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
4	78.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
5	7.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	78.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
7	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
8	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
9	51.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
10	40.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	72.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
2	72.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
3	20.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	72.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
5	20.0	1.0	1.0	2.0	2.0	1.0	2.0	2.0	1.0	2.0
6	20.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
7	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
9	30.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	78.0	1.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0
2	78.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
3	7.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
4	7.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0
5	7.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
6	7.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
7	65.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
8	36.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
9	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
10	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	78.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
2	78.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
3	78.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
4	7.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
5	7.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	7.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
7	38.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	44.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
9	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	28.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	78.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
2	78.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	1.0
3	20.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
4	78.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	1.0
5	20.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
6	20.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
7	56.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	28.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	49.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0